Extra Credit

Due Date: August 23, 2019 by 5:00pm

MAT 022A Linear Algebra

Instructor: Mikhail Gaerlan

Instructions: This assignment is worth 15 extra points added to Midterm 1. The figure below is an electrical circuit diagram. Attached is the section on electrical circuits from *Introductory Linear Algebra: An Applied First Course*. Use Kirchoff's Circuit Laws to set up a system of equations for the currents $I_1, \ldots, I_8$ in each wire. Then use Gaussian Elimination to solve the system. Make sure to write down each row operation used.

Nodes

a. $I_4 = I_1 + I_7$

b. $I_1 + I_2 = I_3$

c. $I_5 = I_2 + I_4 + I_6$

d. $I_3 + I_8 = I_5$

e. $I_6 + I_7 = I_8$

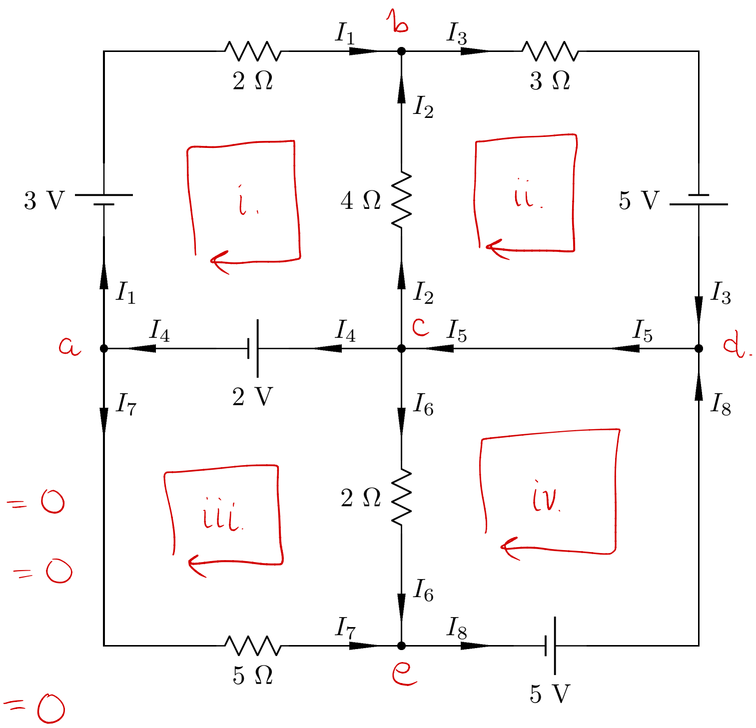
Junctions

i. $3 - 2I_1 + 4I_2 - 2 = 0$

ii. $-4I_2 - 3I_3 + 5 = 0$

iii. $2I_6 - 5 = 0$

iv. $2 - 2I_6 + 5I_7 = 0$



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\begin{array}{c}
 \\ a. \\ b. \\ c. \\ d. \\ e. \\ i. \\ ii. \\ iii. \\ iv.
\end{array}
\begin{array}{c}
\begin{array}{cccccccc} I_1 & I_2 & I_3 & I_4 & I_5 & I_6 & I_7 & I_8 \end{array} \\
\left[\begin{array}{cccccccc|c}
-1 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\
1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & -1 & 1 & -1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & -1 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & 0 \\
-2 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & -4 & -3 & 0 & 0 & 0 & 0 & 0 & -5 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 5 \\
0 & 0 & 0 & 0 & 0 & -2 & 5 & 0 & -2
\end{array}\right]
\end{array}
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\begin{array}{c}
\begin{array}{cccccccc} I_1 & I_2 & I_3 & I_4 & I_5 & I_6 & I_7 & I_8 \end{array} \\
\left[\begin{array}{cccccccc|c}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27}{26} \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7}{26} \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \frac{17}{13} \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \frac{213}{130} \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & \frac{573}{130} \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \frac{5}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \frac{3}{5} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \frac{31}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}\right]
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